



INFO 442

Group 6

Exploratory Data Analysis Report

Investigating the Relationship
Between Classroom Environmental Conditions
and Student Performance

1. Introduction

This EDA investigates how **classroom environmental conditions** (e.g., CO₂, temperature, humidity, and air quality) **relate to student cognitive performance and learning outcomes**. The results will support hypothesis refinement and future predictive modelling.

2. Dataset Overview

The dataset has no missing or duplicate records. It also includes engineered features, such as moving averages, lagged variables, trends, and interactions, to capture temporal patterns and environmental effects.

Table 1. Dataset Summary

Item	Value
Number of Observations	2,000
Number of Variables	28
Numerical Variables	20
Categorical Variables	8
Missing Values	0
Duplicate Records	0
Target Variable	Quiz Score
Memory Usage	437.6 KB

Several high CO₂ observations were retained as valid classroom conditions rather than outliers. Overall, no severe data quality issues were identified that would prevent further analysis.

3. Univariate Analysis

3.1 Distribution of Quiz Scores

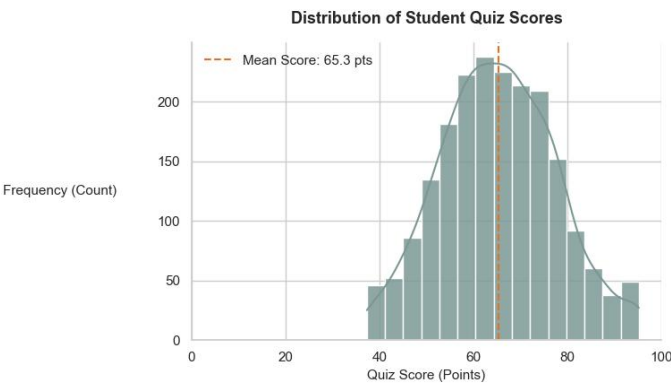


Figure 1. Distribution of Quiz Scores

The histogram illustrates the distribution of quiz scores among students. Quiz scores are approximately normally distributed, indicating sufficient variability in academic performance.

3.2 Distribution of Classroom CO₂ Concentration

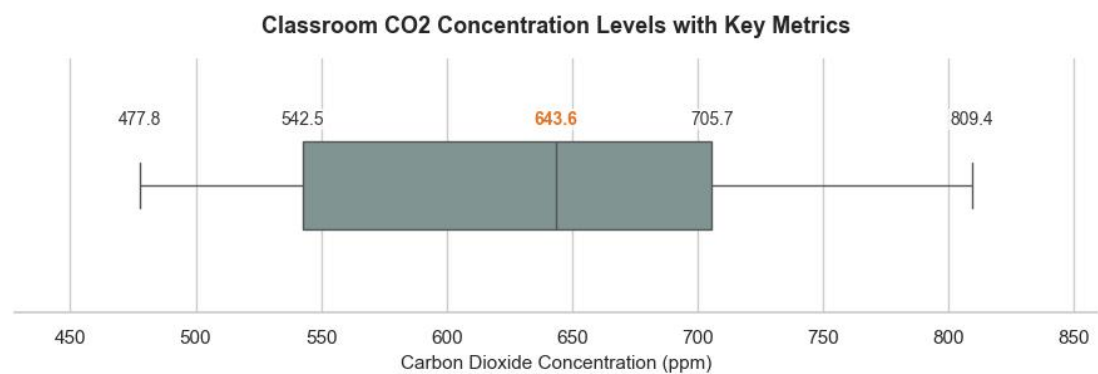


Figure 2. Boxplot of Classroom CO₂ Concentration

The boxplot indicates a wide spread with substantial variation in classroom ventilation conditions. Several observations exceed the upper whisker, indicating periods of poor ventilation. These high-CO₂ events may represent important environmental conditions and were therefore retained for subsequent analyses.

3.3 Distribution of Air Quality Categories

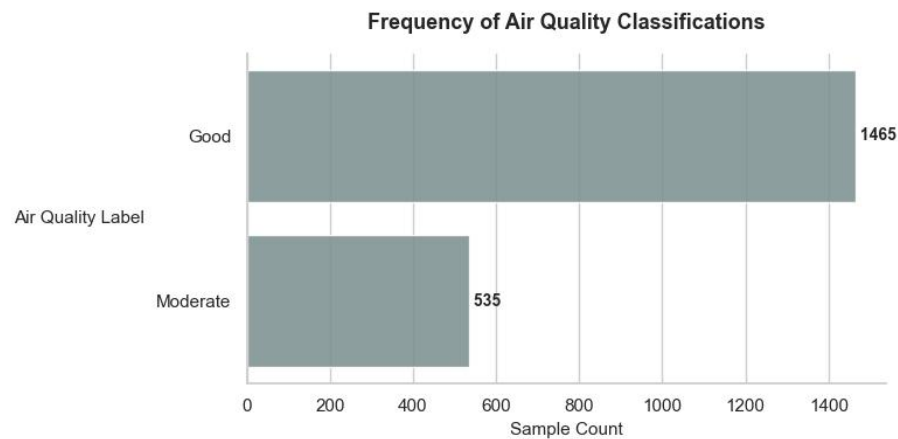


Figure 3. Distribution of Air Quality Categories

The count plot presents the frequency of observations within each air-quality category. Although the categories are imbalanced, both groups contain enough observations for meaningful comparison. This imbalance should be considered during model development because dominant categories may influence predictive performance.

3.4 Distribution of Student Performance Categories

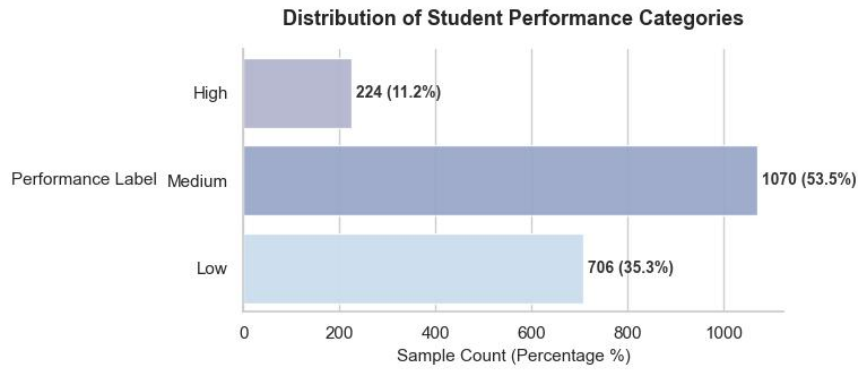


Figure 4. Distribution of Student Performance Labels

The count plot shows the frequency of observations within each performance category. Student performance categories display meaningful variability, suggesting sufficient diversity in academic outcomes.

4. Bivariate Analysis

4.1 CO₂ Concentration and Quiz Performance

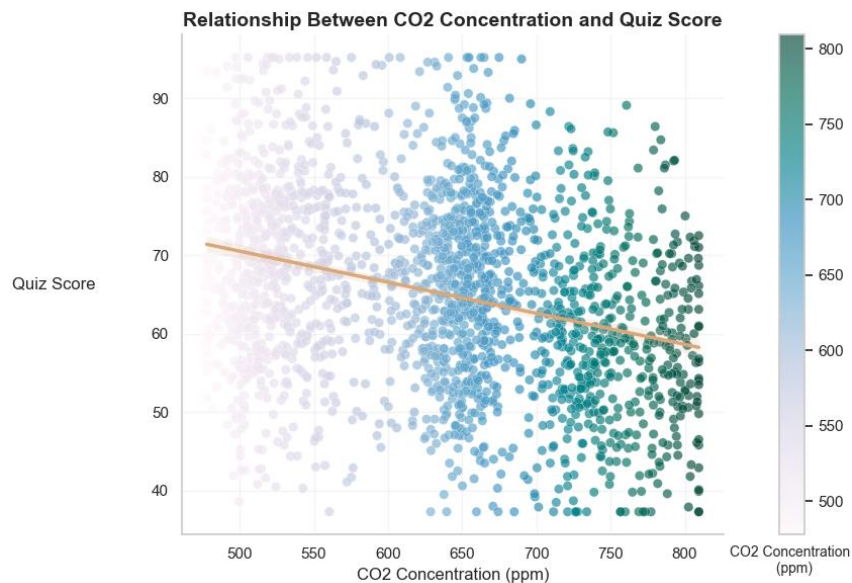


Figure 5. Relationship Between CO₂ Concentration and Quiz Score

The scatterplot illustrates the association between classroom CO₂ concentration and quiz performance.

- ❖ Pearson correlation coefficient: $r = -0.296$
- ❖ $p < 0.001$

A significant **negative relationship** was observed. Students exposed to higher CO₂ levels generally achieved lower quiz scores, supporting the hypothesis that poorer ventilation may be associated with reduced academic performance.

4.2 CO₂ Concentration and Reaction Time

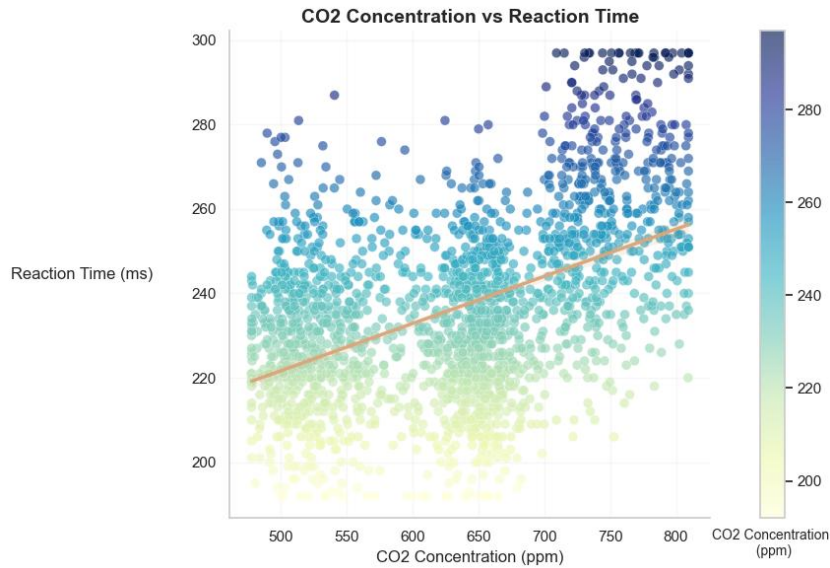


Figure 6. Relationship Between CO₂ Concentration and Reaction Time

The scatterplot shows the relationship between classroom CO₂ concentration and reaction time.

- ❖ Pearson correlation coefficient: $r = 0.471$
- ❖ $p < 0.001$

CO₂ concentration was **positively associated** with reaction time. Students in higher CO₂ environments tended to respond more slowly, indicating reduced cognitive efficiency.

4.3 Air Quality and Academic Performance

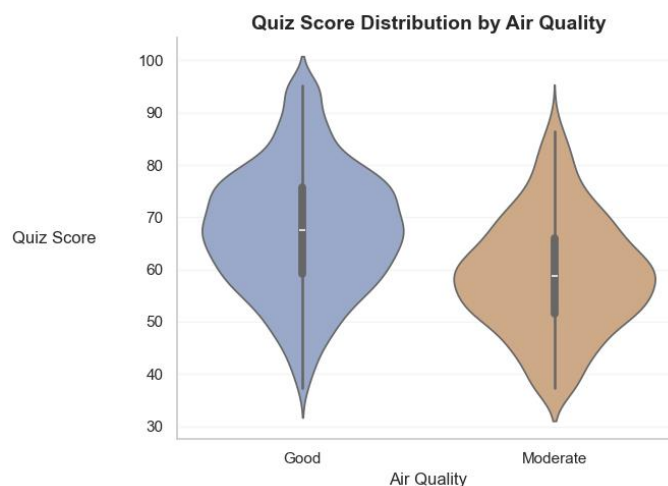


Figure 7. Average Quiz Score Across Air Quality Categories

The plot compares average quiz scores among different air-quality categories. Students in classrooms with good air quality achieved higher quiz scores than those in

moderate air-quality environments. This pattern is consistent with the findings from the CO₂ analysis. **Average quiz scores decline as classroom air quality deteriorates.**

5. Multivariate Analysis

5.1 Correlation Structure

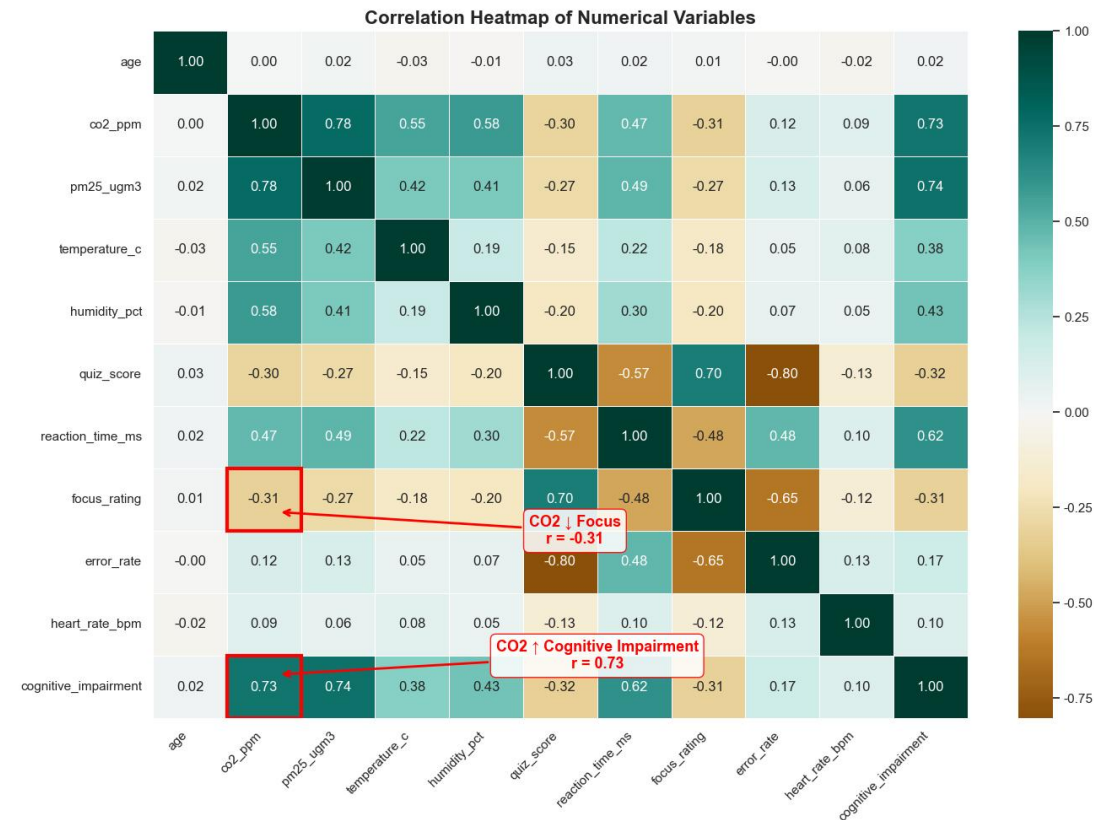


Figure 8. Correlation Heatmap of Numerical Variables

The heatmap summarises pairwise correlations among environmental, cognitive, and academic variables. It shows that **CO₂ concentration** was **negatively associated** with **focus rating** and **positively associated** with **cognitive impairment**. These findings suggest that environmental conditions may affect academic performance through cognitive processes.

Additionally, several variables exhibit moderate intercorrelations, indicating **potential multicollinearity** that should be assessed during predictive modelling.

5.2 Interaction Analysis: Air Quality × Asthma Status

Group	Decline in Quiz Score
Asthma	11.21 points
Non-Asthma	8.16 points

The interaction plot shows that students with asthma exhibited a larger decline in quiz performance under poorer air-quality conditions. Students with asthma experienced a larger decline in quiz score than students without asthma as air quality worsened. This suggests that asthma status may modify the impact of environmental conditions on performance.

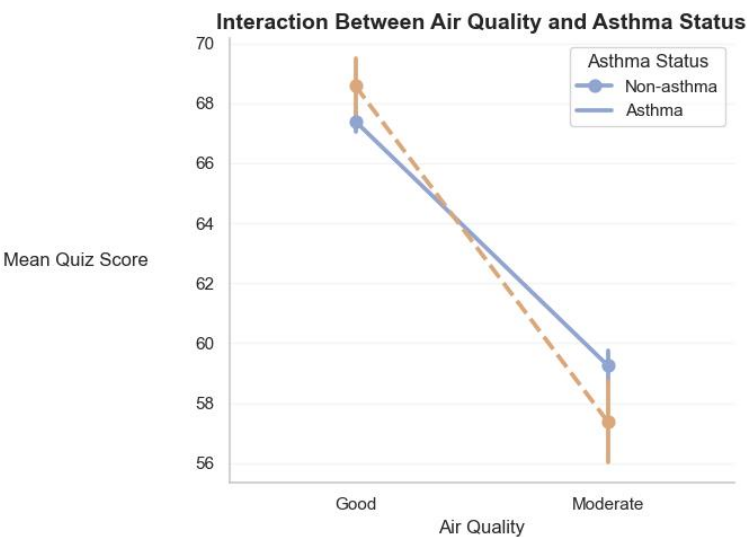


Figure 9. Interaction Between Air Quality and Asthma Status

The observed interaction effect indicates that environmental variables may not operate independently and that **subgroup-specific effects** should be considered during model development.

5.3 Pairwise Relationships Among Variables

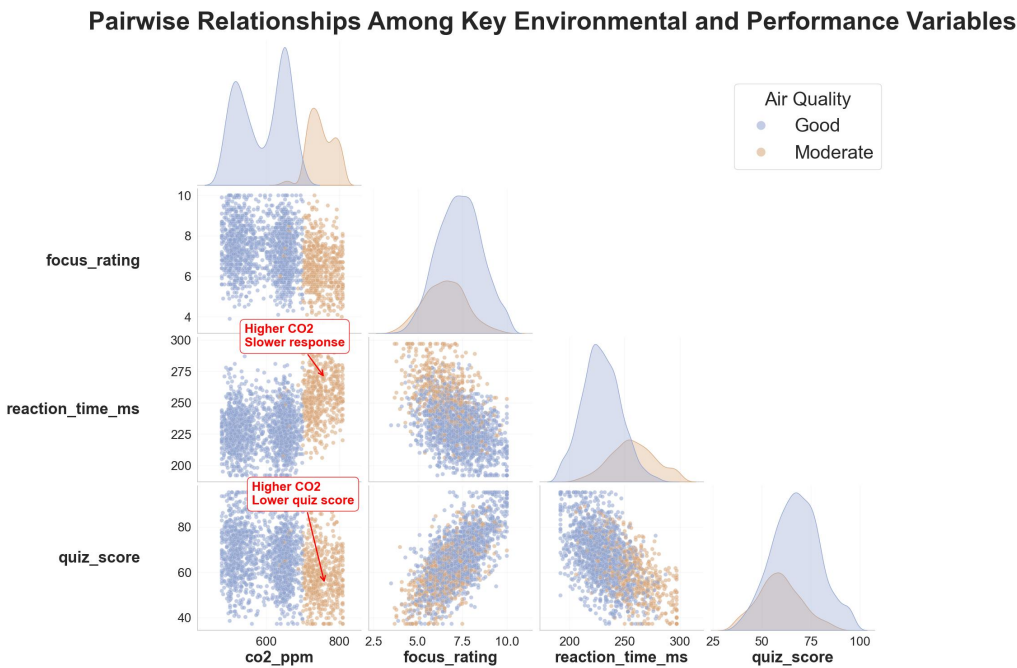


Figure 10. Pairplot of Key Variables

The pairplot confirms several relationships observed previously, including the **negative association between CO₂ concentration and quiz score** and the **positive association between CO₂ concentration and reaction time**. Moreover, several relationships appear **non-linear**, suggesting that tree-based models may capture patterns that linear models fail to detect.

5.4 Interaction Feature Analysis

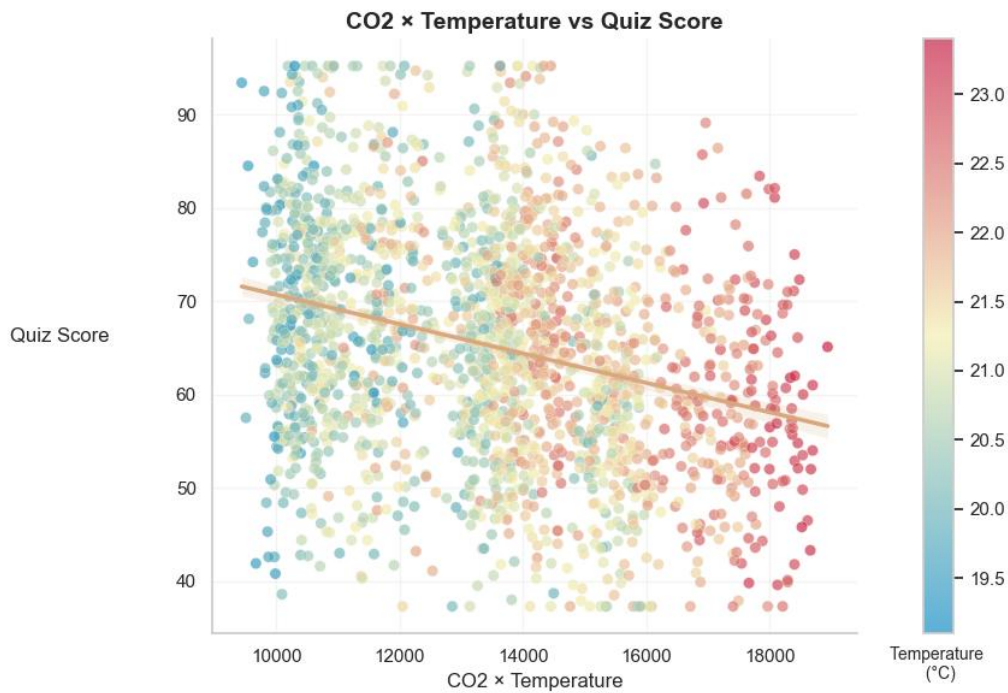


Figure 11. CO₂ × Temperature Interaction and Quiz Score

The colour gradient represents classroom temperature. The interaction feature **CO₂ × Temperature** shows a statistically significant negative association with quiz score ($r = -0.289$, $p = 1.029 \times 10^{-39}$). A similar negative relationship was observed for **CO₂ × Humidity** and quiz score ($r = -0.296$, $p = 8.464 \times 10^{-42}$). These results suggest that combined environmental effects are relevant to student performance and provide additional support for H3.

These interaction terms should be formally tested in the modelling stage to determine whether they improve predictive performance beyond the individual effects of CO₂, temperature, and humidity alone.

6. Revised Hypotheses

H1: Higher classroom CO₂ concentrations are associated with lower quiz scores.

- ❖ **Evidence:** $r = -0.296$; $p < 0.001$
- ❖ **Conclusion:** Supported.

H2: Classroom environmental variables contribute to the prediction of student performance.

- ❖ **Evidence:** CO₂ concentration was significantly associated with both quiz score and reaction time.

- ❖ **Conclusion:** Supported.

H3: The relationship between air quality and student performance is influenced by environmental and health-related interactions.

- ❖ **Evidence:** Interaction analyses (CO₂ × Temperature, CO₂ × Humidity, and Air Quality × Asthma) showed significant associations with performance outcomes.

- ❖ **Conclusion:** Supported.

H4: Students with asthma may experience a greater decline in academic performance under poorer air-quality conditions.

- ❖ **Evidence:** Asthma group decline = 11.21 points; Non-asthma group decline = 8.16 points.

- ❖ **Conclusion:** Supported.

The larger decline observed in students with asthma suggests that health status may act as an effect modifier rather than an independent predictor.

7. Modelling Question

The **primary modelling objective** is to predict student quiz performance using environmental, cognitive, demographic, and interaction variables. The **secondary objective** is to evaluate whether interaction effects provide additional predictive value and improve model performance beyond models based solely on main effects.

8. Conclusion

This exploratory data analysis demonstrates that classroom environmental conditions are meaningfully associated with student cognitive and academic outcomes. Interaction analysis further suggested that students with asthma are more vulnerable to poor classroom environmental conditions.

Overall, the findings support the inclusion of environmental variables in subsequent predictive models and highlight the importance of investigating interaction effects during model development.